

Azure AI Search vs SQL Server 2025 Hyperscale

Native vector + semantic search comparison · cost crossover model · migration map · East US pricing, May 2026

Capability parity matrix

CAPABILITY	AZURE AI SEARCH	SQL SERVER 2025 HYPERSCALE	PARITY
Vector type & distance	HNSW + exhaustive KNN, cosine/dot/Euclidean, up to 4 096 dims , scalar & binary quantization, narrow types (int8, float16)	vector(N) data type up to 1 998 dims , VECTOR_DISTANCE function with cosine/dot/Euclidean. <i>Blocks text-embedding-3-large at 3 072 dims.</i>	Partial
ANN vector index	HNSW, managed automatically per index, GA, vector compression baked in	CREATE VECTOR INDEX ... USING DiskANN — still in public preview as of May 2026. Requires PREVIEW_FEATURES = ON. Exact KNN only for < 50K vectors.	Preview
Lexical / BM25 search	Built-in BM25 with analyzers, synonyms, stop-words, language packs	Full-Text Search (CONTAINS/FREETEXT) + Semantic Search (key phrases / similar docs)	Yes
Hybrid (vector + lexical)	Native Reciprocal Rank Fusion (RRF), single query, single response	Hand-rolled CTE combining FTS rank + VECTOR_DISTANCE with manual RRF	DIY
Semantic re-ranker (L2)	Microsoft-hosted cross-encoder, billed per query,	Not built-in. Requires self-hosted cross-encoder or AOAI re-rank call	Gap

captions + answers +
query rewrite

Integrated vectorization	One-click chunk + embed at index time via AOAI / AI Foundry	sp_invoke_external_rest_endpoint to AOAI, or external pipeline	DIY
Indexers / ingestion connectors	Blob, ADLS, Cosmos, SQL DB, Fabric OneLake, SharePoint, etc. — managed change tracking	ADF, Fabric pipelines, Logic Apps, or custom Functions — manual change tracking	Gap
Document cracking (PDF/DOCX/HTML)	Document Extraction skill + OCR + custom skillsets	Azure AI Document Intelligence called externally; raw text stored in SQL	External
Faceting & filtering	Native facets, filterable fields, scoring profiles	GROUP BY, WHERE, computed/indexed columns, materialized views	Yes
Suggesters / autocomplete	Built-in suggester data structure with fuzzy matching	Trigram or prefix indexes, hand-built	DIY
Geo search	Native Edm.GeographyPoint, geo distance & polygon filters	Native geography / geometry spatial types, spatial indexes	Yes
Agentic retrieval	Native query planner, conversation context, BYOM, billed per reasoning token	Not built-in; orchestrate from app tier with Semantic Kernel / LangChain	Gap
HA / read scale-out	Replicas (1–12), automatic load-balancing	Hyperscale HA replicas (1–4) + named replicas for read scale	Yes

Vertical scale (CPU/RAM)	Fixed per tier — must upgrade tier (S1→S2→S3)	Independent vCore and storage scaling, online vCore changes	Better
Transactional writes	Eventually consistent index updates; no ACID across documents	Full ACID, joins, transactions, foreign keys — same engine as your OLTP	Better
Security: CMK, private link, RBAC	CMK, private endpoints, Entra ID auth, role-based access	TDE w/ CMK, private link, Entra auth, RLS, column encryption	Yes
Backup / PITR	No user-controlled backups (re-index from source)	Continuous PITR, 1–35 day retention, LTR up to 10 years	Better
SLA	99.9% read for ≥2 replicas, 99.9% read/write for ≥3 replicas	99.99% for ≥1 HA replica, 99.995% for zone-redundant HA	Better

Sources: Azure AI Search service limits ([link](#)), Azure SQL Hyperscale docs ([link](#)), SQL Server 2025 vectors ([link](#)).